

1s, 2s, 9s and 10s Complement

The 9's complement of 546700 is $999999 - 546700 = 453299$.

The 9's complement of 012398 is $999999 - 012398 = 987601$.

the 10's complement of 012398 is 987602

the 10's complement of 246700 is 753300

The 1's complement of 1011000 is 0100111.

The 1's complement of 0101101 is 1010010.

the 2's complement of 1101100 is 0010100

the 2's complement of 0110111 is 1001001

Forming the Two's Complement

- Negative numbers are stored in two's complement notation
- Represents the **additive Inverse**

Starting value	00000001
Step 1: reverse the bits	11111110
Step 2: add 1 to the value from Step 1	11111110 +00000001
Sum: two's complement representation	11111111

Note that $00000001 + 11111111 = 00000000$

Binary Subtraction

- When subtracting $A - B$, convert B to its two's complement
- Add A to $(-B)$

$$\begin{array}{r} 00001100 \\ - 00000011 \\ \hline \end{array} \longrightarrow \begin{array}{r} 00001100 \\ 11111101 \\ \hline 00001001 \end{array}$$

Practice: Subtract 0101 from 1001.

Subtraction using Complements

Using 10's complement, subtract $72532 - 3250$.

$$\begin{array}{r} M = 72532 \\ 10\text{'s complement of } N = + \underline{96750} \\ \text{Sum} = 169282 \\ \text{Discard end carry } 10^5 = -\underline{100000} \\ \text{Answer} = 69282 \end{array}$$

Note that M has five digits and N has only four digits. Both numbers must have the same number of digits, so we write N as 03250. Taking the 10's complement of N produces a 9 in the most significant position. The occurrence of the end carry signifies that $M \geq N$ and that the result is therefore positive.

Using 10's complement, subtract $3250 - 72532$.

$$\begin{array}{r} M = 03250 \\ 10\text{'s complement of } N = +\underline{27468} \\ \text{Sum} = 30718 \end{array}$$

There is no end carry. Therefore, the answer is $-(10\text{'s complement of } 30718) = -69282$.

Note that since $3250 < 72532$, the result is negative. Because we are dealing with unsigned

Subtraction using Complements

Given the two binary numbers $X = 1010100$ and $Y = 1000011$, perform the subtraction (a) $X - Y$ and (b) $Y - X$ by using 2's complements.

$$\begin{array}{rcl} \text{(a)} & X = & 1010100 \\ & 2\text{'s complement of } Y = + & \underline{0111101} \\ & \text{Sum} = & 10010001 \\ & \text{Discard end carry } 2^7 = - & \underline{10000000} \\ & \text{Answer: } X - Y = & 0010001 \end{array}$$

$$\begin{array}{rcl} \text{(b)} & Y = & 1000011 \\ & 2\text{'s complement of } X = + & \underline{0101100} \\ & \text{Sum} = & 1101111 \end{array}$$

There is no end carry. Therefore, the answer is $Y - X = -(2\text{'s complement of } 1101111) = -0010001$.

Learn How To Do the Following:

- Form the two's complement of a hexadecimal integer
- Convert signed binary to decimal
- Convert signed decimal to binary
- Convert signed decimal to hexadecimal
- Convert signed hexadecimal to decimal

BCD

Table 1.4
Binary-Coded Decimal (BCD)

Decimal Symbol	BCD Digit
0	0000
1	0001
2	0010
3	0011
4	0100
5	0101
6	0110
7	0111
8	1000
9	1001

4	0100	4	0100	8	1000
+5	+0101	+8	+1000	+9	1001
9	1001	12	1100	17	10001
			+0110		+0110
			10010		10111

BCD	1	1		
	0001	1000	0100	184
	+0101	0111	0110	+576
Binary sum	0111	10000	1010	
Add 6		0110	0110	
BCD sum	0111	0110	0000	760

$$(185)_{10} = (0001\ 1000\ 0101)_{BCD} = (10111001)_2$$

BCD Addition Example

Represent the unsigned decimal numbers 842 and 537 in BCD, and then show the steps necessary to form their sum.

1000	0100	0010	(842)
<u>0101</u>	<u>0011</u>	<u>0111</u>	(+537)
1101	0111	1001	
<u>0110</u>			
0001	0011	0111	0101 (1,379)

Other Decimal Codes

Table 1.5
Four Different Binary Codes for the Decimal Digits

Decimal Digit	BCD 8421	2421	Excess-3	8, 4, -2, -1
0	0000	0000	0011	0000
1	0001	0001	0100	0111
2	0010	0010	0101	0110
3	0011	0011	0110	0101
4	0100	0100	0111	0100
5	0101	1011	1000	1011
6	0110	1100	1001	1010
7	0111	1101	1010	1001
8	1000	1110	1011	1000
9	1001	1111	1100	1111
Unused bit combinations	1010	0101	0000	0001
	1011	0110	0001	0010
	1100	0111	0010	0011
	1101	1000	1101	1100
	1110	1001	1110	1101
	1111	1010	1111	1110

Character Storage

- Character sets
 - Standard ASCII (0 – 127)
 - Extended ASCII (0 – 255)
 - ANSI (0 – 255)
 - Unicode (0 – 65,535)
- Null-terminated String
 - Array of characters followed by a *null byte*
- Using the ASCII table
 - back inside cover of book

Numeric Data Representation

- pure binary
 - can be calculated directly
- ASCII binary
 - string of digits: "01010101"
- ASCII decimal
 - string of digits: "65"
- ASCII hexadecimal
 - string of digits: "9C"

next: Boolean Operations

ASCII

Table 1.7
American Standard Code for Information Interchange (ASCII)

$b_4b_3b_2b_1$	$b_7b_6b_5$							
	000	001	010	011	100	101	110	111
0000	NUL	DLE	SP	0	@	P	`	p
0001	SOH	DC1	!	1	A	Q	a	q
0010	STX	DC2	"	2	B	R	b	r
0011	ETX	DC3	#	3	C	S	c	s
0100	EOT	DC4	\$	4	D	T	d	t
0101	ENQ	NAK	%	5	E	U	e	u
0110	ACK	SYN	&	6	F	V	f	v
0111	BEL	ETB	.	7	G	W	g	w
1000	BS	CAN	(	8	H	X	h	x
1001	HT	EM	)	9	I	Y	i	y
1010	LF	SUB	*	:	J	Z	j	z
1011	VT	ESC	+	;	K	[	k	{
1100	FF	FS	.	<	L	\	l	
1101	CR	GS	-	=	M	]	m	}
1110	SO	RS	.	>	N	^	n	~
1111	SI	US	/	?	O	_	o	DEL

ASCII

Control characters

NUL	Null	DLE	Data-link escape
SOH	Start of heading	DC1	Device control 1
STX	Start of text	DC2	Device control 2
ETX	End of text	DC3	Device control 3
EOT	End of transmission	DC4	Device control 4
ENQ	Enquiry	NAK	Negative acknowledge
ACK	Acknowledge	SYN	Synchronous idle
BEL	Bell	ETB	End-of-transmission block
BS	Backspace	CAN	Cancel
HT	Horizontal tab	EM	End of medium
LF	Line feed	SUB	Substitute
VT	Vertical tab	ESC	Escape
FF	Form feed	FS	File separator
CR	Carriage return	GS	Group separator
SO	Shift out	RS	Record separator
SI	Shift in	US	Unit separator
SP	Space	DEL	Delete

3. What is the decimal representation of each of the following unsigned binary integers?

a. 11111000

b. 11001010

c. 11110000

3. (a) 248 (b) 202 (c) 240

6. What is the sum of each pair of binary integers?

a. 10101111 + 11011011

b. 10010111 + 11111111

c. 01110101 + 10101100

6. (a) 110001010 (b) 110010110 (c) 100100001

12. What is the hexadecimal representation of each of the following binary numbers?

a. 0011 0101 1101 1010

b. 1100 1110 1010 0011

c. 1111 1110 1101 1011

12. (a) 35DA (b) CEA3 (c) FEDB

16. What is the unsigned decimal representation of each hexadecimal integer?

a. 62

b. 1C9

c. 6A5B

16. (a) 98 (b) 457 (c) 27227

18. What is the 16-bit hexadecimal representation of each signed decimal integer?

a. -32

b. -62

18. (a) FFE0 (b) FFC2

22. What is the decimal representation of the following signed binary numbers?

a. 10000000

b. 11001100

c. 10110111

22. (a) -128 (b) -52 (c) -73

20. The following 16-bit hexadecimal numbers represent signed integers. Convert to decimal.

a. 7F9B

b. 8230

20. (a) +32667 (b) -32208

24. What is the 8-bit binary (two's complement) representation of each of the following signed decimal integers?

a. -72

b. -98

c. -26

24. (a) 10111000 (b) 10011110 (c) 11100110

What is the largest value you can represent using a 256-bit *unsigned* integer?

$$2^{256} - 1$$